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Assi Ghat Boat Electrification Study-Cell Level

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Introduction

The previous electrification study conducted for the Assi Ghat boat fleet established the feasibility of replacing internal combustion engines with electric propulsion systems and provided high-level battery bank capacities suitable for three major boat categories. While that report validated operating costs, energy sizing, and payback period, it treated the battery bank as a single conceptual unit.

This continuation study delves deeper into the internal structure of these battery banks, breaking them down into individual lithium cells and modules to understand the engineering, reliability, cost, and lifecycle challenges that emerge at the component level.

Background

Electric propulsion systems rely heavily on reliable, long-lasting battery storage. The earlier study recommended battery capacities of:

- 30 kWh for standard (10 HP) boats
- 60 kWh for mid-sized (20 HP) boats
- 400 kWh for large passenger cruisers (180 HP)

However, these values alone are insufficient for engineering implementation. By transitioning from the bank perspective to the cell perspective, we can better evaluate:

- Component cost breakdown
- Thermal management and safety
- Battery longevity
- Maintenance requirements
- Manufacturing quality risks

This cell-level perspective is critical for safe and scalable deployment along the Ganga riverfront.

Lithium Cell Selection for Marine Applications

The preferred technology for this purpose is **Lithium Iron Phosphate ($LiFePO_4$)**, **26800 cylindrical cell** due to:

- High thermal stability
- Resistance to fire/thermal runaway
- Long cycle life (3,000–5,000+ cycles ≈ 8–10+ years)
- Low maintenance
- A commonly available unit cell typically has:
- **Nominal voltage: 3.6 V**
- **Capacity: 6 Ah**
- **Weight : 130 g**

These cells represent the smallest electrical “building block” used in marine battery banks.

Construction of Modules and Banks

Module Formation (Series Connection)

Voltage is increased by connecting 3.6 V cells in series:

- 30 cells × 3.6 V = 108 V
- Capacity remains 6 Ah
- Energy in one module = 108 V × 6 Ah = .648 kWh
- This 108 V module forms the standardized power block used across various boat types.

Bank Formation (Parallel Connection)

Multiple .648 kWh modules are connected in parallel to reach the required total storage capacity. This increases energy (kWh), while voltage stays constant.

Cell-Level Configuration Per Boat Type

Boat type	Bank Capacity	Module Size	Required Modules	Individual Cells	Weight
Standard (10 HP)	30 KWh	.648 kWh	50 modules (≈32.4 kWh)	1500 cells (30 * 50)	195 kgs
Mid sized (20 HP)	60 KWh	.648 kWh	100 modules (≈64.8 kWh)	3000 cells (30 * 100)	390 kgs
Large (180 HP)	400 KWh	.648 kWh	620 modules (≈401.7 kWh)	18600 cells (30 * 620)	2418 kgs

- Note: The large boat would likely use a higher voltage system (e.g., 400V-800V), but this would still require a similar total number of individual cells, just arranged differently.

Density Caclulations (Wh/kg)

The weight of one cell of 26800 cylindrical cell is 13

energy density of one cell is volt * current /weight
= 3.6* 6 /.130
=167 wh/kg

Hence the Energy density of the cell is 167 Wh/kg

Critical Cell-Level Challenges

Battery Management System (BMS)

- Acts as the central control “brain” of the battery pack.
- Must monitor voltage and temperature of all 105 individual cells.
- Detects hazardous conditions such as overheating or cell failure.
- Can shut down the entire pack to prevent damage and safety risks.

Cell Balancing and Matching

- Cells must be factory-graded and matched for capacity, age, and resistance.
- Poor-quality or mismatched cells become unbalanced during charging cycles.
- Results in some cells overcharging and others undercharging.
- Can lead to rapid degradation or failure of the entire ₹4,20,000 battery bank.

Second Life & Recycling

- $LiFePO_4$ cells have an 8–10+ year marine service life.
- After retirement, modules still retain usable capacity.
- Can be repurposed for stationary storage at charging stations on the ghats.
- Supports a circular energy economy and reduces overall battery waste.

CONCLUSION

Breaking the battery bank down into its individual cells reveals that:

- Technical feasibility remains strong
- BMS design is the most critical safety factor
- Proper cell matching is essential for longevity
- Financial estimates remain valid at component level
- Long-term sustainability benefits can be enhanced through second-life usage

For successful adoption, pilot programs must emphasize real-world heat testing, certified sourcing, and strict quality control standards. Structured support mechanisms from government bodies will significantly accelerate fleet-wide electrification.